

Willian Vieira PhD

Data scientist with a PhD in quantitative ecology, specializing in forecasting, Bayesian hierarchical modeling, causal inference, and production ML pipelines for large-scale longitudinal and observational data. I build reproducible systems that move models from messy real-world data to uncertainty-aware decisions.

DATA SCIENCE & ENGINEER EXPERIENCE

2025
|
2024
(1 yr 9 m)

- Data Analyst**
Habitat, Montreal, Canada
Architected end-to-end R/Python ETL and prediction pipelines to deploy ML models into production environments | Built and maintained reproducible pipelines processing large-scale, multi-source datasets for continental-scale analytics and forecasting | Designed a cloud-based lakehouse architecture on Azure to support scalable feature retrieval, model training, and downstream analytics | Implemented metadata-driven workflows and automated validation to improve training-dataset governance and pipeline reliability | Led the transition to a Unix-based production environment using Docker, CI/CD, automated setup scripts, and DevOps best practices | Translated client and project needs into clear analytical outputs for non-technical stakeholders

2024
|
2017

- PhD Research**
Integrative Ecology Lab, Sherbrooke, Canada
Built Bayesian hierarchical models to forecast forest dynamics from noisy, biased, continental-scale climate and inventory datasets | Developed reproducible HPC workflows () to run thousands of simulations for forecast evaluation and uncertainty propagation | Developed open-source software libraries (, ) for spatial demographic modelling with version control, automated testing, and reproducible workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (, )

2022
|
2020
(part-time)

- Biostatistician**
Environment and Climate Change Canada - Quebec, Canada
Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **geospatial pipeline** with version-controlled workflows, automated documentation, and analytical outputs designed for long-term stakeholder reference

EDUCATION

2024
|
2017

- PhD, Ecology**
Université de Sherbrooke - Sherbrooke, Canada
 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016
|
2015

- Masters 2, Agroecology and Resource Management**
Bordeaux Sciences Agro, Bordeaux, France
 *Modelling the dispersion of weed species in agricultural landscapes*

2015
|
2010

- BSc in Agronomy**
Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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 [LinkedIn/WillVieira](#)
 [GitHub/WillVieira](#)
 [Publications](#)

SKILLS

Machine Learning & Modelling:
Forecasting · Bayesian inference · Hierarchical models · Causal inference · A/B testing · Experimental design · Forecast evaluation · Uncertainty

Programming & Database:
R · Python · SQL · Stan · Julia · DuckDB · Bash

Data Engineering & Cloud:
ETL/ELT · Production ML pipelines · Feature engineering · Azure · AWS · Docker · Apache Arrow · Parquet · HPC

DevOps & Reproducibility:
Git · CI/CD · Unix · Unit testing · Make · Quarto · Jupyter · Automated validation

OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students
· Taught +250 students across biology, engineering, and graduate programs (programming, statistics, ecology) · **5 published** papers & preprints with ~100 citations · Translated client needs into technical analyses and delivered clear, actionable results

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional

CV source code hosted on 

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